

Amogh Patki | College Park, MD

UX Engineer | Software Engineer | Full Stack Developer | Game Developer
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EDUCATION

Master of Science in Human-Computer Interaction

Expected May 2028

University of Maryland | College Park, MD

Bachelor of Science in Software Engineering

2025

Western Governors University | Salt Lake City, UT

Relevant Coursework: Algorithms & Data Structures, Technical Writing, Cloud Foundations (AWS), Advanced Data Management, Network & Security, Back-End Programming

Associate of Applied Science, Computer Game Development

Mt. Hood Community College | Gresham, OR

Relevant Coursework: Interface Design for Interactive Applications

TECHNICAL SKILLS

Languages: JavaScript, Rust, C#, Java

Web/Frameworks: React, TypeScript, Vite, Angular, Spring Boot, HTML5

Cloud & DevOps: Amazon Web Services, Git

Databases: PostgreSQL, Supabase

Tools: Bevy, Unity, Godot Engine, Photoshop

Certifications: AWS Certified Cloud Practitioner, CompTIA Project+, ITIL 4 Foundation

EXPERIENCE

Founding Engineer @Algobrick.ai

May 2026 to September 2026

Construction takeoff and estimation platform (React, TypeScript, Vite, Supabase, AWS)

- Rebuilt the core drawing tools (counts, lines, and areas) by benchmarking against Bluebeam, the industry-standard takeoff software, and observing an experienced user's drawing workflow to find missing interactions.
- Turned frequent actions buried in Bluebeam's menus into single gestures, such as double-click to add points with alignment guides and one-step cutouts on areas, saving up to 3 clicks per action.
- Built the takeoff workspace UI, including a label panel, a takeoff table, and automatic quantity totals for approved elements, with scale and approval requirements that keep incomplete or incorrect takeoffs out of later steps.
- Linked the drawing canvas and takeoff table both ways, so selecting an element in either one highlights or centers it in the other, speeding up in-app navigation and removing confusion over which shape matched which entry.
- Designed the element edit menu to grow with an evolving product, separating shared actions (visibility, approval, relabeling) from element-specific ones (joining lines) so future developers could add tools without reworking the layout.
- Built the first version of the estimation table, with a pipeline that auto-fills vendor quotes and rates from the company's Excel pricing sheet, replacing manual price entry for every line item.
- Built self-serve sign-up and login on Supabase with Google and Microsoft SSO, required email confirmation, password reset, and automatic user profile creation.
- Built a "contact the founder" feature using AWS SNS, giving users a direct line to leadership for feedback and support.